

Battery Degradation Datasets

Download links for the MSc EV Engineering dissertation: *Duty Cycle-Aware Battery Degradation Analytics for EV Fleet Applications*

Usha Rani V. | University of East London | Compiled 21 June 2026

Core Battery Degradation Datasets

1. NASA Prognostics (PCoE) Battery Dataset

Li-ion 18650 charge/discharge cycling, impedance (EIS) measurements, and capacity-fade trajectories under controlled accelerated ageing. Run-to-failure time series, ideal for SoH, RUL and degradation modelling.

Official repository (NASA PCoE):

<https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/>

NASA Open Data Portal (landing + download):

<https://data.nasa.gov/dataset/li-ion-battery-aging-datasets>

Direct ZIP mirror (Battery Data Set):

<https://phm-datasets.s3.amazonaws.com/NASA/5.+Battery+Data+Set.zip>

Note: NASA has periodically migrated this archive; if the direct ZIP is temporarily offline, use the Open Data Portal landing page above.

2. CALCE Battery Dataset (University of Maryland)

Center for Advanced Life Cycle Engineering open test data: continuous full and partial cycling, storage, dynamic driving profiles, OCV and impedance. Form factors cylindrical / pouch / prismatic; chemistries LCO, LFP, NMC. The CS2 / CX2 series is the most widely used for SoH and RUL work.

Main download page (all series + readme + citations):

<https://calce.umd.edu/battery-data>

3. University of Oxford Battery Degradation Dataset

Long-term cycling of 8 Kokam 740 mAh pouch cells (Dataset 1), using a CC charge + ARTEMIS urban drive-cycle discharge, with characterisation every 100 cycles. Directly relevant to your drive-cycle / duty-cycle framing. Hosted on the Oxford Research Archive (ORA), Open Database Licence.

Oxford Battery Degradation Dataset 1 (ORA record + download):

<https://ora.ox.ac.uk/objects/uuid:03ba4b01-cfed-46d3-9b1a-7d4a7bdf6fac>

Battery Intelligence Lab hub (all Oxford datasets, one page):

<https://battery-intelligence-lab.github.io/data-and-code/>

Related Oxford sets on ORA (optional, for path-dependence / energy trading):

<https://ora.ox.ac.uk/objects/uuid:de62b5d2-6154-426d-bcbb-30253ddb7d1e>

<https://ora.ox.ac.uk/objects/uuid:9aae61af-2949-49f1-8ad5-6aea448979e5>

Quick reference table

Dataset	Host	Best link
NASA PCoE	data.nasa.gov	data.nasa.gov/dataset/li-ion-battery-aging-datasets
CALCE	calce.umd.edu	calce.umd.edu/battery-data
Oxford	ora.ox.ac.uk	ora.ox.ac.uk/.../03ba4b01

All three are open-access for research. CALCE and Oxford ask that any publication using the data cites their original papers (citation details are on each download page). Drive-cycle profiles (WLTP, UDDS, FTP-75, HWFET, NEDC) are separate from these and come from EPA / UNECE / standards sources, not the battery archives above.